

The Normal Distribution

ANSWER KEY



Properties of the normal distribution

- 1 State three key properties of a normal distribution's shape.

A normal distribution is: (1) symmetric and bell-shaped, centered at its mean; (2) unimodal (a single peak at the mean); (3) its total area under the curve equals 1, and it never touches the horizontal axis (it extends infinitely in both directions, getting closer and closer to zero).

- 2 Explain what the two parameters μ and σ control in a normal distribution's shape.

μ (the mean) controls the location of the center of the distribution (where the peak is). σ (the standard deviation) controls the spread: a larger σ produces a wider, flatter curve, while a smaller σ produces a narrower, taller curve — but both remain symmetric and bell-shaped.

The 68-95-99.7 rule

- 3 Adult male heights are approximately normal with mean 70 inches and standard deviation 3 inches. Using the 68-95-99.7 rule, find the range of heights containing the middle 95% of adult men.

The 95% range spans $\mu \pm 2\sigma = 70 \pm 2(3) = 70 \pm 6$, so from 64 to 76 inches.

- 4 Using the same distribution, find the approximate percentage of adult men taller than 76 inches.

76 inches is $\mu + 2\sigma$. By the 68-95-99.7 rule, 95% of the data lies within 2 standard deviations, leaving 5% split evenly in the two tails: 2.5% above 76 inches and 2.5% below 64 inches.

Standardizing and using the normal distribution

- 5 Using the same height distribution (mean 70, standard deviation 3), find the z-score for a height of 77.5 inches, and use a standard normal table value of $P(Z < 2.5) \approx 0.9938$ to find the proportion of men shorter than 77.5 inches.

$z = \frac{77.5 - 70}{3} = \frac{7.5}{3} = 2.5$. $P(X < 77.5) = P(Z < 2.5) \approx 0.9938$, or about 99.38% of men are shorter than 77.5 inches.

- 6 Find the proportion of men between 65 and 75 inches tall, given $P(Z < 1.67) \approx 0.9525$ and $P(Z < -1.67) \approx 0.0475$.

$z_{65} = \frac{65 - 70}{3} \approx -1.67$. $z_{75} = \frac{75 - 70}{3} \approx 1.67$.
 $P(65 < X < 75) = P(Z < 1.67) - P(Z < -1.67) \approx 0.9525 - 0.0475 = 0.905$, or about 90.5%.

Visualizing the standard normal distribution

- 7 The graph shows the standard normal distribution ($\mu = 0$, $\sigma = 1$), with the area to the left of $z = 1$ shaded (though shading itself is not shown, imagine it visually).

a Using $P(Z < 1) \approx 0.8413$, find the area under the curve to the right of $z = 1$.
 $P(Z > 1) = 1 - P(Z < 1) = 1 - 0.8413 = 0.1587$.

b Explain, using the curve's symmetry, why $P(Z < -1) = P(Z > 1)$. The standard normal curve is symmetric about $z = 0$. By symmetry, the area to the left of -1 is a mirror image of the area to the right of 1 , so $P(Z < -1)$ and $P(Z > 1)$ must be equal.

Working backward: finding values from percentiles

- 8 SAT scores are approximately normal with mean 1050 and standard deviation 150. A student scores at the 90th percentile, corresponding to $z \approx 1.28$. Find that student's approximate score.

$$x = \mu + z\sigma = 1050 + 1.28(150) = 1050 + 192 = 1242.$$

- 9 Using the same distribution, find the score that separates the bottom 10% of scores, given that $z \approx -1.28$ corresponds to the 10th percentile.

$$x = \mu + z\sigma = 1050 + (-1.28)(150) = 1050 - 192 = 858.$$

Assessing normality

- 10 Explain how a normal probability plot (or normal quantile plot) can be used to assess whether a data set is approximately normal, and what a roughly straight-line pattern indicates.

A normal probability plot compares the ordered data values to what would be expected if the data were exactly normal. If the plotted points fall close to a straight line, the data is approximately normal. Systematic curvature or bending away from a straight line (such as an S-shape or clear bends at the ends) suggests the data deviates from normality — for example, skewness or heavy tails.

Applying the normal distribution to real data

- 11 Pregnancy lengths are approximately normal with mean $\mu = 266$ days and standard deviation $\sigma = 16$ days. Find the z -score for a pregnancy lasting 250 days.

$$z = \frac{250 - 266}{16} = \frac{-16}{16} = -1.$$

- 12 Using the same pregnancy-length distribution and the 68-95-99.7 rule, find the approximate percentage of pregnancies lasting between 234 and 298 days.

$234 = 266 - 32 = \mu - 2\sigma$ and $298 = 266 + 32 = \mu + 2\sigma$, so this range spans $\mu \pm 2\sigma$. By the 68-95-99.7 rule, approximately 95% of pregnancies fall in this range.

Combining two independent normal random variables

- 13 A vending machine dispenses a soda whose volume is normally distributed with mean 355 mL and standard deviation 4 mL, independently of a snack bag whose weight is normally distributed with mean 50 g and standard deviation 2 g. If T represents the combined "total" formed by adding the soda volume (in mL) and snack weight (in g) as a single number, find the mean of T . (This is a simplified illustrative combination, not a real physical unit.)

For independent normal random variables, means add directly: $\mu_T = 355 + 50 = 405$.

- 14 Battery lifespans are normally distributed with mean 500 hours and standard deviation 40 hours. Find the z-score for a battery lasting 460 hours, and describe its relative position.

$z = \frac{460 - 500}{40} = \frac{-40}{40} = -1$. This battery's lifespan is exactly one standard deviation below the mean.

- 15 A company manufactures resistors with resistance values approximately normal, mean 100 ohms and standard deviation 5 ohms. A resistor is considered defective if its resistance is below 90 ohms or above 110 ohms. Using the 68-95-99.7 rule, estimate the percentage of resistors that are defective, and explain your reasoning step by step.

$90 = 100 - 2(5) = \mu - 2\sigma$ and $110 = 100 + 2(5) = \mu + 2\sigma$, so the acceptable range spans $\mu \pm 2\sigma$. By the 68-95-99.7 rule, approximately 95% of resistors fall within this range (not defective), leaving approximately 5% defective (split as roughly 2.5% below 90 ohms and 2.5% above 110 ohms).

- 16 Two students are comparing their standardized test scores from different tests. Student X scored 78 on a test with mean 70 and standard deviation 8. Student Y scored 165 on a different test with mean 150 and standard deviation 12. Using Z-scores, determine who performed relatively better compared to their own test's distribution of test-takers, showing full reasoning.

Student X: $z = \frac{78 - 70}{8} = \frac{8}{8} = 1$. Student Y: $z = \frac{165 - 150}{12} = \frac{15}{12} = 1.25$. Since $1.25 > 1$, Student Y performed relatively better compared to their own test's distribution of test-takers, even though the raw score gap looks different for each test.

Multiple choice — easy 15

- 17** A normal distribution has mean 64 and standard deviation 9. Using the 68-95-99.7 rule, find the range containing the middle 95% of values.

a. (46, 82) ✓

b. (37, 91)

c. (46, 81)

d. (47, 82)

The middle 95% range spans $\mu \pm 2\sigma = 64 \pm 2(9) = 64 \pm 18$, so from 46 to 82: (46, 82).

- 18** A normal distribution has mean 162 and standard deviation 13. Using the 68-95-99.7 rule, find the range containing the middle 68% of values.

a. (136, 188)

b. (149, 174)

c. (149, 175) ✓

d. (150, 175)

The middle 68% range spans $\mu \pm 1\sigma = 162 \pm 1(13) = 162 \pm 13$, so from 149 to 175: (149, 175).

- 19** A normal distribution has mean 332 and standard deviation 16. Using the 68-95-99.7 rule, find the range containing the middle 99.7% of values.

a. (268, 396)

b. (284, 379)

c. (285, 380)

d. (284, 380) ✓

The middle 99.7% range spans $\mu \pm 3\sigma = 332 \pm 3(16) = 332 \pm 48$, so from 284 to 380: (284, 380).

- 20** A normal distribution has mean 228 and standard deviation 4. Using the 68-95-99.7 rule, find the range containing the middle 68% of values.
- a. (225, 232)
 - b. (220, 236)
 - c. (224, 232) ✓
 - d. (224, 231)

The middle 68% range spans $\mu \pm 1\sigma = 228 \pm 1(4) = 228 \pm 4$, so from 224 to 232: (224, 232).

- 21** A normal distribution has mean 457 and standard deviation 11. Using the 68-95-99.7 rule, find the range containing the middle 99.7% of values.
- a. (425, 490)
 - b. (424, 490) ✓
 - c. (424, 489)
 - d. (413, 501)

The middle 99.7% range spans $\mu \pm 3\sigma = 457 \pm 3(11) = 457 \pm 33$, so from 424 to 490: (424, 490).

- 22** A normal distribution has mean 195 and standard deviation 12. Using the 68-95-99.7 rule, find the range containing the middle 99.7% of values.
- a. (159, 230)
 - b. (160, 231)
 - c. (159, 231) ✓
 - d. (147, 243)

The middle 99.7% range spans $\mu \pm 3\sigma = 195 \pm 3(12) = 195 \pm 36$, so from 159 to 231: (159, 231).

- 23** A normal distribution has mean 50 and standard deviation 14. Using the 68-95-99.7 rule, find the range containing the middle 68% of values.

- a. (22, 78)
- b. (36, 64) ✓
- c. (36, 63)
- d. (37, 64)

The middle 68% range spans $\mu \pm 1\sigma = 50 \pm 1(14) = 50 \pm 14$, so from 36 to 64: (36, 64).

- 24** A normal distribution has mean 288 and standard deviation 6. Using the 68-95-99.7 rule, find the range containing the middle 68% of values.

- a. (283, 294)
- b. (282, 294) ✓
- c. (276, 300)
- d. (282, 293)

The middle 68% range spans $\mu \pm 1\sigma = 288 \pm 1(6) = 288 \pm 6$, so from 282 to 294: (282, 294).

- 25** A normal distribution has mean 438 and standard deviation 35. Using the 68-95-99.7 rule, find the range containing the middle 95% of values.

- a. (368, 508) ✓
- b. (368, 507)
- c. (333, 543)
- d. (369, 508)

The middle 95% range spans $\mu \pm 2\sigma = 438 \pm 2(35) = 438 \pm 70$, so from 368 to 508: (368, 508).

- 26** A normal distribution has mean 138 and standard deviation 40. Using the 68-95-99.7 rule, find the range containing the middle 95% of values.

a. (58, 218) ✓

b. (18, 258)

c. (58, 217)

d. (59, 218)

The middle 95% range spans $\mu \pm 2\sigma = 138 \pm 2(40) = 138 \pm 80$, so from 58 to 218: (58, 218).

- 27** A normal distribution has mean 481 and standard deviation 20. Using the 68-95-99.7 rule, find the range containing the middle 99.7% of values.

a. (421, 541) ✓

b. (422, 541)

c. (421, 540)

d. (401, 561)

The middle 99.7% range spans $\mu \pm 3\sigma = 481 \pm 3(20) = 481 \pm 60$, so from 421 to 541: (421, 541).

- 28** A normal distribution has mean 88 and standard deviation 18. Using the 68-95-99.7 rule, find the range containing the middle 68% of values.

a. (70, 106) ✓

b. (70, 105)

c. (52, 124)

d. (71, 106)

The middle 68% range spans $\mu \pm 1\sigma = 88 \pm 1(18) = 88 \pm 18$, so from 70 to 106: (70, 106).

- 29 A normal distribution has mean 64 and standard deviation 15. Using the 68-95-99.7 rule, find the range containing the middle 95% of values.
- a. (35, 94)
 - b. (19, 109)
 - c. (34, 93)
 - d. (34, 94) ✓

The middle 95% range spans $\mu \pm 2\sigma = 64 \pm 2(15) = 64 \pm 30$, so from 34 to 94: (34, 94).

- 30 A normal distribution has mean 361 and standard deviation 35. Using the 68-95-99.7 rule, find the range containing the middle 95% of values.
- a. (291, 430)
 - b. (256, 466)
 - c. (292, 431)
 - d. (291, 431) ✓

The middle 95% range spans $\mu \pm 2\sigma = 361 \pm 2(35) = 361 \pm 70$, so from 291 to 431: (291, 431).

- 31 A normal distribution has mean 414 and standard deviation 13. Using the 68-95-99.7 rule, find the range containing the middle 95% of values.
- a. (388, 439)
 - b. (389, 440)
 - c. (375, 453)
 - d. (388, 440) ✓

The middle 95% range spans $\mu \pm 2\sigma = 414 \pm 2(13) = 414 \pm 26$, so from 388 to 440: (388, 440).

Multiple choice — medium 15

- 32 A normal distribution has mean $\mu = 101$ and standard deviation $\sigma = 4$. Find the approximate percentage of values below $x = 103$, to two decimal places.

a. 69.15% ✓

b. 72.15%

c. 66.15%

d. 30.85%

$z = \frac{103 - 101}{4} = 0.50$. Using the standard normal distribution, $P(Z < 0.50) \approx 0.6915$, or about 69.15%.

- 33 A normal distribution has mean $\mu = 94$ and standard deviation $\sigma = 2$. Find the approximate percentage of values below $x = 89$, to two decimal places.

a. 99.38%

b. 2.62%

c. 3.62%

d. 0.62% ✓

$z = \frac{89 - 94}{2} = -2.50$. Using the standard normal distribution, $P(Z < -2.50) \approx 0.0062$, or about 0.62%.

- 34 A normal distribution has mean $\mu = 365$ and standard deviation $\sigma = 18$. Find the approximate percentage of values below $x = 356$, to two decimal places.

a. 30.85% ✓

b. 27.85%

c. 69.15%

d. 33.85%

$z = \frac{356 - 365}{18} = -0.50$. Using the standard normal distribution, $P(Z < -0.50) \approx 0.3085$, or about 30.85%.

- 35 A normal distribution has mean $\mu = 64$ and standard deviation $\sigma = 10$. Find the approximate percentage of values below $x = 39$, to two decimal places.

a. 0.62% ✓

b. 2.62%

c. 3.62%

d. 99.38%

$z = \frac{39 - 64}{10} = -2.50$. Using the standard normal distribution, $P(Z < -2.50) \approx 0.0062$, or about 0.62%.

- 36 A normal distribution has mean $\mu = 160$ and standard deviation $\sigma = 16$. Find the approximate percentage of values below $x = 184$, to two decimal places.

a. 90.32%

b. 96.32%

c. 93.32% ✓

d. 6.68%

$z = \frac{184 - 160}{16} = 1.50$. Using the standard normal distribution, $P(Z < 1.50) \approx 0.9332$, or about 93.32%.

- 37 A normal distribution has mean $\mu = 477$ and standard deviation $\sigma = 20$. Find the approximate percentage of values below $x = 527$, to two decimal places.

a. 96.38%

b. 0.62%

c. 97.38%

d. 99.38% ✓

$z = \frac{527 - 477}{20} = 2.50$. Using the standard normal distribution, $P(Z < 2.50) \approx 0.9938$, or about 99.38%.

- 38 A normal distribution has mean $\mu = 485$ and standard deviation $\sigma = 10$. Find the approximate percentage of values below $x = 505$, to two decimal places.

- a. 94.72%
- b. 2.28%
- c. 95.72%
- d. 97.72% ✓

$z = \frac{505 - 485}{10} = 2$. Using the standard normal distribution, $P(Z < 2) \approx 0.9772$, or about 97.72%.

- 39 A normal distribution has mean $\mu = 174$ and standard deviation $\sigma = 16$. Find the approximate percentage of values below $x = 198$, to two decimal places.

- a. 6.68%
- b. 90.32%
- c. 93.32% ✓
- d. 96.32%

$z = \frac{198 - 174}{16} = 1.50$. Using the standard normal distribution, $P(Z < 1.50) \approx 0.9332$, or about 93.32%.

- 40 A normal distribution has mean $\mu = 110$ and standard deviation $\sigma = 16$. Find the approximate percentage of values below $x = 126$, to two decimal places.

- a. 15.87%
- b. 81.13%
- c. 87.13%
- d. 84.13% ✓

$z = \frac{126 - 110}{16} = 1$. Using the standard normal distribution, $P(Z < 1) \approx 0.8413$, or about 84.13%.

- 41 A normal distribution has mean $\mu = 454$ and standard deviation $\sigma = 6$. Find the approximate percentage of values below $x = 457$, to two decimal places.

- a. 72.15%
- b. 69.15% ✓
- c. 66.15%
- d. 30.85%

$z = \frac{457 - 454}{6} = 0.50$. Using the standard normal distribution, $P(Z < 0.50) \approx 0.6915$, or about 69.15%.

- 42 A normal distribution has mean $\mu = 338$ and standard deviation $\sigma = 20$. Find the approximate percentage of values below $x = 388$, to two decimal places.

- a. 99.38% ✓
- b. 96.38%
- c. 0.62%
- d. 97.38%

$z = \frac{388 - 338}{20} = 2.50$. Using the standard normal distribution, $P(Z < 2.50) \approx 0.9938$, or about 99.38%.

- 43 A normal distribution has mean $\mu = 268$ and standard deviation $\sigma = 4$. Find the approximate percentage of values below $x = 280$, to two decimal places.

- a. 96.87%
- b. 99.87% ✓
- c. 97.87%
- d. 0.13%

$z = \frac{280 - 268}{4} = 3$. Using the standard normal distribution, $P(Z < 3) \approx 0.9987$, or about 99.87%.

- 44 A normal distribution has mean $\mu = 99$ and standard deviation $\sigma = 2$. Find the approximate percentage of values below $x = 104$, to two decimal places.

a. 99.38% ✓

b. 97.38%

c. 96.38%

d. 0.62%

$z = \frac{104 - 99}{2} = 2.50$. Using the standard normal distribution, $P(Z < 2.50) \approx 0.9938$, or about 99.38%.

- 45 A normal distribution has mean $\mu = 318$ and standard deviation $\sigma = 16$. Find the approximate percentage of values below $x = 342$, to two decimal places.

a. 90.32%

b. 6.68%

c. 96.32%

d. 93.32% ✓

$z = \frac{342 - 318}{16} = 1.50$. Using the standard normal distribution, $P(Z < 1.50) \approx 0.9332$, or about 93.32%.

- 46 A normal distribution has mean $\mu = 343$ and standard deviation $\sigma = 4$. Find the approximate percentage of values below $x = 335$, to two decimal places.

a. 5.28%

b. 2.28% ✓

c. 4.28%

d. 97.72%

$z = \frac{335 - 343}{4} = -2$. Using the standard normal distribution, $P(Z < -2) \approx 0.0228$, or about 2.28%.

Multiple choice — hard 15

- 47 A normal distribution has mean $\mu = 144$ and standard deviation $\sigma = 16$. Find the approximate percentage of values between $x_1 = 136$ and $x_2 = 152$, to two decimal places.

- a. 69.15%
- b. 34.29%
- c. 42.29%
- d. 38.29% ✓

$$z_1 = \frac{136 - 144}{16} = -0.50. z_2 = \frac{152 - 144}{16} = 0.50. P(-0.50 < Z < 0.50) \approx 0.3829, \text{ or about } 38.29\%$$

- 48 A normal distribution has mean $\mu = 237$ and standard deviation $\sigma = 2$. Find the approximate percentage of values between $x_1 = 233$ and $x_2 = 240$, to two decimal places.

- a. 91.04% ✓
- b. 87.04%
- c. 95.04%
- d. 93.32%

$$z_1 = \frac{233 - 237}{2} = -2. z_2 = \frac{240 - 237}{2} = 1.50. P(-2 < Z < 1.50) \approx 0.9104, \text{ or about } 91.04\%.$$

- 49 A normal distribution has mean $\mu = 57$ and standard deviation $\sigma = 4$. Find the approximate percentage of values between $x_1 = 63$ and $x_2 = 69$, to two decimal places.

- a. 10.55%
- b. 2.55%
- c. 99.87%
- d. 6.55% ✓

$$z_1 = \frac{63 - 57}{4} = 1.50. z_2 = \frac{69 - 57}{4} = 3. P(1.50 < Z < 3) \approx 0.0655, \text{ or about } 6.55\%.$$

- 50 A normal distribution has mean $\mu = 354$ and standard deviation $\sigma = 18$. Find the approximate percentage of values between $x_1 = 318$ and $x_2 = 363$, to two decimal places.

a. 66.87% ✓

b. 69.15%

c. 70.87%

d. 62.87%

$$z_1 = \frac{318 - 354}{18} = -2. \quad z_2 = \frac{363 - 354}{18} = 0.50. \quad P(-2 < Z < 0.50) \approx 0.6687, \text{ or about } 66.87\%.$$

- 51 A normal distribution has mean $\mu = 123$ and standard deviation $\sigma = 10$. Find the approximate percentage of values between $x_1 = 133$ and $x_2 = 148$, to two decimal places.

a. 11.24%

b. 19.24%

c. 99.38%

d. 15.24% ✓

$$z_1 = \frac{133 - 123}{10} = 1. \quad z_2 = \frac{148 - 123}{10} = 2.50. \quad P(1 < Z < 2.50) \approx 0.1524, \text{ or about } 15.24\%.$$

- 52 Scores on an exam are approximately normal with mean $\mu = 390$ and standard deviation $\sigma = 15$. A student scores at the 95th percentile, corresponding to $z \approx 1.645$. Find that student's approximate score.

a. 365

b. 405

c. 425

d. 415 ✓

$$x = \mu + z\sigma = 390 + (1.645)(15) = 415.$$

- 53 Scores on an exam are approximately normal with mean $\mu = 837$ and standard deviation $\sigma = 50$. A student scores at the 95th percentile, corresponding to $z \approx 1.645$. Find that student's approximate score.
- a. 909
 - b. 755
 - c. 919 ✓
 - d. 929

$$x = \mu + z\sigma = 837 + (1.645)(50) = 919.$$

- 54 Scores on an exam are approximately normal with mean $\mu = 399$ and standard deviation $\sigma = 10$. A student scores at the 95th percentile, corresponding to $z \approx 1.645$. Find that student's approximate score.
- a. 405
 - b. 415 ✓
 - c. 425
 - d. 383

$$x = \mu + z\sigma = 399 + (1.645)(10) = 415.$$

- 55 Scores on an exam are approximately normal with mean $\mu = 295$ and standard deviation $\sigma = 20$. A student scores at the 10th percentile, corresponding to $z \approx -1.28$. Find that student's approximate score.
- a. 279
 - b. 259
 - c. 321
 - d. 269 ✓

$$x = \mu + z\sigma = 295 + (-1.28)(20) = 269.$$

- 56 Scores on an exam are approximately normal with mean $\mu = 762$ and standard deviation $\sigma = 50$. A student scores at the 90th percentile, corresponding to $z \approx 1.28$. Find that student's approximate score.
- a. 826 ✓
 - b. 836
 - c. 698
 - d. 816

$$x = \mu + z\sigma = 762 + (1.28)(50) = 826.$$

- 57 A company manufactures parts with a measurement approximately normal, mean 152 and standard deviation 5. A part is considered defective if its measurement is below 142 or above 162 (i.e. outside $\mu \pm 2\sigma$). Using the 68-95-99.7 rule, estimate the percentage of parts that are defective.
- a. 0.3%
 - b. 5% ✓
 - c. 32%
 - d. 2.5%

$142 = \mu - 2\sigma$ and $162 = \mu + 2\sigma$, so the acceptable range spans $\mu \pm 2\sigma$. By the 68-95-99.7 rule, approximately 95% of parts fall within this range, leaving approximately 5% defective (roughly 2.5% in each tail).

- 58 A company manufactures parts with a measurement approximately normal, mean 224 and standard deviation 27. A part is considered defective if its measurement is below 170 or above 278 (i.e. outside $\mu \pm 2\sigma$). Using the 68-95-99.7 rule, estimate the percentage of parts that are defective.
- a. 32%
 - b. 2.5%
 - c. 5% ✓
 - d. 0.3%

$170 = \mu - 2\sigma$ and $278 = \mu + 2\sigma$, so the acceptable range spans $\mu \pm 2\sigma$. By the 68-95-99.7 rule, approximately 95% of parts fall within this range, leaving approximately 5% defective (roughly 2.5% in each tail).

- 59 A company manufactures parts with a measurement approximately normal, mean 85 and standard deviation 28. A part is considered defective if its measurement is below 1 or above 169 (i.e. outside $\mu \pm 3\sigma$). Using the 68-95-99.7 rule, estimate the percentage of parts that are defective.

a. 0.3% ✓

b. 0.15%

c. 5%

d. 32%

$1 = \mu - 3\sigma$ and $169 = \mu + 3\sigma$, so the acceptable range spans $\mu \pm 3\sigma$. By the 68-95-99.7 rule, approximately 99.7% of parts fall within this range, leaving approximately 0.3% defective (roughly 0.15% in each tail).

- 60 A company manufactures parts with a measurement approximately normal, mean 84 and standard deviation 15. A part is considered defective if its measurement is below 39 or above 129 (i.e. outside $\mu \pm 3\sigma$). Using the 68-95-99.7 rule, estimate the percentage of parts that are defective.

a. 0.3% ✓

b. 5%

c. 0.15%

d. 32%

$39 = \mu - 3\sigma$ and $129 = \mu + 3\sigma$, so the acceptable range spans $\mu \pm 3\sigma$. By the 68-95-99.7 rule, approximately 99.7% of parts fall within this range, leaving approximately 0.3% defective (roughly 0.15% in each tail).

- 61 A company manufactures parts with a measurement approximately normal, mean 22 and standard deviation 30. A part is considered defective if its measurement is below -8 or above 52 (i.e. outside $\mu \pm 1\sigma$). Using the 68-95-99.7 rule, estimate the percentage of parts that are defective.

a. 32% ✓

b. 16%

c. 0.3%

d. 5%

$-8 = \mu - 1\sigma$ and $52 = \mu + 1\sigma$, so the acceptable range spans $\mu \pm 1\sigma$. By the 68-95-99.7 rule, approximately 68% of parts fall within this range, leaving approximately 32% defective (roughly 16% in each tail).